Classification Using KNN and Naive Bayes

Full Python Program

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score

X, y = load_iris(return_X_y=True)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42)

knn = KNeighborsClassifier(n_neighbors=5)
knn.fit(X_train, y_train)
print("k-NN accuracy:", format(accuracy_score(y_test, knn.predict(X_test)), ".2f"))

nb = GaussianNB()
nb.fit(X_train, y_train)
print("Naive Bayes accuracy:", format(accuracy_score(y_test, nb.predict(X_test)), ".2f"))
```

Full Output

```
k-NN accuracy: 1.00
Naive Bayes accuracy: 0.98
```

Requirement: Install scikit-learn before running the program: `pip install scikit-learn`

Run the program in Python IDLE by saving the file and pressing F5.